

Harsh Jannawar

San Francisco, California, United States | hjannawar014@gmail.com | (425) 780-8648 | github.com/Harshj143 | linkedin.com/in/harsh-jannawar

PROFESSIONAL SUMMARY

Security Engineer who builds secure, resilient systems and ships security tooling rather than just finding problems. Experienced across application security, detection engineering, cloud, and AI/LLM security, aligned with OWASP, MITRE ATT&CK, and NIST.

EDUCATION

University of Washington

Master of Science, *Cybersecurity Engineering*

Sep 2024 - Jun 2026

GPA: 3.93/4

Symbiosis Skills and Professional University

Bachelor of Technology, *CSIT (Cybersecurity)*

Aug 2020 - May 2024

GPA: 3.65/4

SKILLS

Technical Skills: Security tool development, Threat modeling, Application security reviews, Penetration testing, Exploit development, Secure code review, Authentication and authorization

Programming Languages: Python, JavaScript, Node.js, Bash Scripting, SQL

Frameworks & Libraries: ISO 27001/42001, NIST, SOC 2, asyncio, YAML

Databases: MongoDB, Postgres, SQL

Tools & Software: Burp Suite, Metasploit, Nmap, Nessus, Nuclei, OWASP ZAP, IDA Pro, Splunk, Docker, Git, CloudWatch, GuardDuty, Amazon ECR, Amazon ECS, AWS Lambda, S3, CloudTrail, WAF

Cloud Platforms: AWS, Azure, Oracle Cloud Infrastructure

WORK EXPERIENCE

SecureAls | AI Cybersecurity Engineering Intern | California

Jun 2025 - Aug 2025

- Developed a PII detection and redaction pipeline using Presidio, regex-based validation, and packet analysis to protect sensitive data across 10k test samples, improving sensitive data protection accuracy from 65% to 95%.
- Conducted adversarial testing and secure code reviews across AI platforms to uncover 20+ critical vulnerabilities including authentication bypass, prompt injection, and token leakage prior to production release.
- Integrated SAST, DAST, dependency scanning, and secret detection into CI/CD pipelines to provide continuous security feedback during development, reducing security review cycles by 60%.
- Built and tested end-to-end authentication flows across engineering services, including secure login, session handling, token validation, role-based access control, and protected API routes to reduce unauthorized access risk across application workflows.

SecureThings | Security Analyst Intern | Pune, India

Jun 2023 - May 2024

- Built an event-driven scanning platform using Dockerized asynchronous Python, Nmap, Nuclei, and more workflows deployed via Amazon ECR and ECS, saved in S3 and MongoDB, triggered by AWS Lambda on new user request, cutting asset analysis time from 9+ hours to under 30 minutes.
- Performed application and infrastructure security assessments using Nessus, Nmap, and Burp Suite to triage 15+ high-severity vulnerabilities with validated exploits.
- Designed backend risk-scoring logic to prioritize findings based on exploitability, exposed services, affected technologies, severity, and business context for faster remediation decisions.
- Strengthened AWS and Azure environments by improving IAM permissions, container security controls, secrets handling, and exposed service configurations, remediating 25+ cloud and application security gaps.

Pune Metro Rail Project | Security Intern | Pune, India

Jun 2022 - Jul 2022

- Reviewed IAM roles and security group configurations across 20+ cloud resources to identify over-permissive access and exposed services, recommending least-privilege changes that reduced the attack surface across 15+ findings.

PROJECTS

Inline Proxy - Security Gateway for AI Agents |

Jun 2026 - Present

- Built a transparent MCP proxy that enforces role-aware, per-tool policy (allow/block/redact/rewrite/quarantine/human-approval) on every agent tool call - pluggable asyncio pipeline, layered YAML policy packs, fail-closed by default, 200+ tests in CI.
- Added defense-in-depth controls that block prompt-injection exfiltration: PII/secret redaction (checksum-validated detectors + tokenization vault), taint tracking, per-session risk scoring with auto-suspend, and an LLM behavioral anomaly monitor.
- Roadmap (designed & documented): connector packs for GitHub/Jira/Slack, OIDC identity via Okta/Auth0, policy-as-code CI/CD, and audit streaming to Splunk/S3 (SIEM).

Cocktail.exe - Malware & Reverse Engineering

- Built a Windows malware sample with persistence, startup manipulation, and obfuscation (UPX, Base64), then reverse-engineered a separate sample in x86 assembly and produced sandbox-based threat-intel reports correlating behavior into IOCs and detection strategies.

Buffer Overflow Exploit Development - SEED Labs

- Exploited a stack-based buffer overflow by disabling ASLR and overwriting the return address at offset 116 with a NOP-sled shellcode payload to spawn a root reverse shell; documented StackGuard canary countermeasures.

Linux Kernel Firewall - SEED Labs

- Built a kernel-level packet-filtering firewall in C using Netfilter hooks (PRE_ROUTING through POST_ROUTING), enforcing drop rules for ICMP and Telnet traffic verified via live kernel logs.

CERTIFICATIONS & ACHIEVEMENTS

- CompTIA Security+ - CompTIA
- CompTIA Pentest+ - CompTIA
- TCM Practical Ethical Hacking - TCM
- Oracle Cloud Infrastructure Foundations - Oracle
- Certified LLM Security Expert - RTL
- Ranked Top 1% on TryHackMe - TryHackMe
- Won Capture The Flag Competition - UWB GreyHats
- Published Cybersecurity & Technical Blogs - Medium